9.2 Populations, Samples, and Hypotheses

Name _____ Period ____ Date _____

Algebra II 9.2 · Student sheet

I can…

- tell the difference between a population and a sample, and between a parameter and a statistic.
- use a simulation to find proportions.
- use a simulation to decide whether to reject a hypothesis.

Part 1 · Is the coin fair?

A friend got 9 heads in 10 flips and says the coin is lucky. Flip a fair coin 10 times. Heads: _____

Heads	0–3	4	5	6	7	8	9–10
Class count							

How often did a fair coin give 9 or 10 heads? _____ Is 9 heads unusual for a fair coin? _____

Part 2 · Notes: fill in the blanks

Population: _____. **Sample:** _____.

A **parameter** describes a _____; a **statistic** describes a _____.

Reject a hypothesis when the result is very _____ if the hypothesis is true.

Part 3 · Examples (work them with the class)

Example 1

A school surveys 100 of its 1,200 students about the lunch menu. Identify the population and the sample.

Example 2

Tell whether each number is a parameter or a statistic.

a. The mean age of all 30 members of a club is 34.

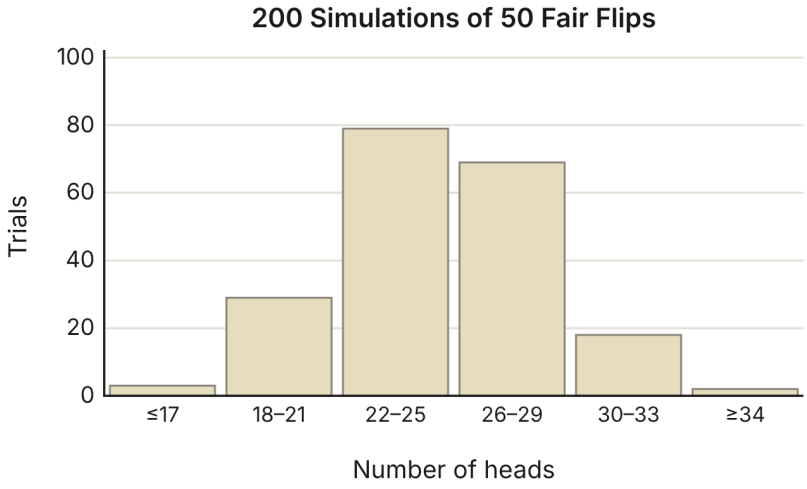
b. In a survey of 1,000 voters, 52% plan to vote yes.

Example 3

A spinner is spun 50 times and lands on red 18 times. What proportion of the spins landed on red?

Example 4

You claim a coin is fair. In 50 flips, you get 34 heads. A simulation of 200 trials of 50 fair coin flips gave 34 or more heads in only 2 trials. What should you conclude?



Part 4 · Partner practice

Partner 1

Tell whether each data set is a population or a sample.

a. the ages of every person counted in a town's census

b. the heights of 30 randomly chosen 10th graders

Partner 2

Tell whether each number is a parameter or a statistic.

a. A poll of 500 shoppers finds that 40% prefer store A.

b. The 25 players on a team average 6 feet tall.